Korea Technology and Information Promotion Agency for SMEs

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Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**Cyber-Resilient Framework for Networked Robotic Systems: Advanced Detection and Mitigation of Geometric Transformation Attacks**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display(), Secondary Battery( ), Advanced Mobility( ), Next-gen Nuclear Power( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity (√), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Information、Communication		Mechanical、Materials	
	Medium Category	Small Category	Medium Category	Small Category
	Information Security	Network System Security	Robotics/ Automation Machinery	Robot Control and Intelligence Technologies
Concept of Proposed Technology	The proposed technology develops a **Cyber-Resilient Robotic Framework** designed to detect and neutralize "perfectly undetectable" affine transformation-based attacks. Unlike traditional cybersecurity that monitors data packets, this system uses **Lie group theory and differential geometry** to monitor the physical-mathematical consistency of robotic motion. By implementing nonlinear **State Monitoring Signature Functions (SMSFs)** along with advanced **homomorphic encryption (HE)** algorithms, the technology ensures that any malicious attempt to alter a robot's physical trajectory—even those that appear mathematically consistent to standard controllers—is immediately identified and mitigated.			

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	Key features: • **Geometric Attack Detection:** Uses Lie group-based modeling to identify potential exploits in which potent "invisible" may be implemented in mechanical systems. • **Nonlinear Signature Functions:** Implementation of SMSFs that create mathematical "incompatibilities" for attackers, making it impossible for them to alter physical behavior without breaking signature consistency. • **Homomorphically Encrypted Control and Attack detection**: Leverages advanced HE schemes to limit attack vectors, while achieving end-to-end security for networked systems. • **Multi-Platform Resiliency:** Validated defense mechanisms for teleoperation (remote surgery/hazardous handling), mobile robotics (logistics), and industrial manipulators (manufacturing).
Rationale for Joint R&D Project	As Korea advances its smart factory initiatives and Industry 4.0 transformation, networked robotic systems are increasingly deployed across manufacturing and automation sectors. However, current cybersecurity solutions that focus solely on communication-layer protection leave critical vulnerabilities in the control architecture itself, making systems susceptible to sophisticated geometric attacks that traditional methods cannot identify. This R&D project addresses these vulnerabilities by developing defense mechanisms that monitor physical-mathematical consistency of robotic motion. The collaboration leverages the PI's expertise in geometric control theory and encrypted control systems with Korean SME partners' practical knowledge of industrial robotics deployment. This synergy enables: • Development of security-enhanced robotic control architectures applicable to Korean manufacturing environments. • Integration of state-of-the-art, end-to-end secure and private control methodologies into existing robotic systems. • Enabling AI based attack intent predictions for advanced forensics of cyberattacks • Establishment of cybersecurity standards for next-generation robotic systems in Korea
Key Words	Cybersecurity, Robotic Systems, False Data Injection Attacks, Lie Group Theory, State Monitoring Signature Functions, Homomorphic encryption, Networked Control Systems, Geometric Transformations, Industrial Automation

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL) [2]	Start of the Project	Level (2)	End of the Project	Level (5)
R&D Objectives	**Aim 1: Geometric Vulnerability Analysis** Develop a comprehensive theoretical framework for analyzing perfectly undetectable false data injection and transformation attacks across both linear and nonlinear robotic systems. Characterize how geometric symmetries in system dynamics enable attacks through coordinate transformations using Lie group theory and differential geometry. **Aim 2: State Monitoring Signature Functions** Introduce state monitoring signature functions as a novel countermeasure against perfectly undetectable attacks. Develop synthesis methods for constructing nonlinear state combinations that deliberately violate the geometric symmetries exploited by attackers, ensuring mathematical incompatibilities between plant dynamics and monitoring outputs. The framework will incorporate machine learning algorithms for real-time attack classification and parameter estimation, enabling adaptive detection thresholds and intelligent identification of attack patterns across scaling, reflection, and shear transformations. **Aim 3: Comprehensive Validation** Validate theoretical frameworks and countermeasures through experimental evaluation on robotic testbeds including industrial manipulators and mobile/domestic robotic platforms.			
R&D Target Specifications	• Attack Detection Performance: 99% detection rate for scaling, reflection, and shear attacks; <1% false positive rate • Real-time Performance: <50ms end-to-end latency for control and detection operations • Control Performance Under Attack: Maintain performance within 5% of nominal metrics for 90% of operation time • Attack Estimation Resistance: Require ≥1000 samples along diverse trajectories for adversarial estimation • Platform Coverage: Validated on mobile robotic platforms (including muti-agent systems) and industrial manipulators			
R&D Plan	Year 1: Theoretical Foundation and Platform Setup • Develop mathematical framework for geometric attack vulnerability analysis • Derive conditions for perfectly undetectable attacks in linear and nonlinear systems • Set up mobile robot and manipulator testbeds Year 2: Countermeasure Development and Implementation • Design and implement state monitoring signature functions • Develop synthesis methods for transformation-resistant monitoring • Implement attack scenarios across all three platforms • Deploy secure control architecture with integrated monitoring Year 3: Validation and Technology Transfer • Conduct comprehensive validation experiments on all platforms • Verify performance metrics and detection capabilities • Develop implementation guidelines for Korean SME partners • Release open-source tools and security assessment framework			

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• Apply geometric attack vulnerability analysis using Lie group theory and differential geometry • Design and implement state monitoring signature functions for attack detection • Provide experimental testbeds including mobile robot platform and industrial manipulator • Provide technical training and knowledge transfer to Korean SME partners • Support integration of security mechanisms into Korean industrial robotic systems
Expected R&R of Korean SME Partner	• Identify critical security requirements for Korean manufacturing and automation applications • Provide industrial robotics expertise and real-world deployment scenarios • Assist in adapting theoretical frameworks to practical industrial control systems • Facilitate field testing in actual Korean manufacturing environments • Contribute to development of implementation guidelines and best practices • Support commercialization pathway for security-enhanced robotic systems

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5. Expected Outcomes/Impact

Technical Outcomes:

This R&D project will deliver a comprehensive cyber-resilient robotic framework capable of detecting and neutralizing perfectly undetectable geometric transformation attacks. This framework will advance the state of the art in secure robotic control by providing mathematical guarantees for attack detection while maintaining real-time performance requirements. The project will produce validated countermeasures implemented across three representative robotic platforms: bilateral teleoperation systems, mobile robots, and industrial manipulators.

Industrial Impact:

From an industrial perspective, this project will provide Korean SME partners with cutting-edge cybersecurity technologies and system integration expertise applicable to next-generation robotic manufacturing systems. The outcomes will support technology commercialization and product development to enable secure and resilient robotic solutions for industrial automation, remote operations, and domestic applications. Korean SMEs will gain competitive advantages through:

- Enhanced security capabilities for robotic products and services
- Reduced vulnerability to sophisticated cyberattacks in networked automation systems
- Implementation expertise for security-aware control architectures
- Access to advanced detection and monitoring tools

Broader Impact:

The project is expected to strengthen the competitiveness of Korean robotics SMEs and contribute to the growth of the secure AI-driven robotics industry through international collaboration. The project will establish new paradigms for designing inherently secure robotic systems and create educational materials for training the next generation of robotics engineers in cybersecurity principles. Expected deliverables include:

- Open-source software tools for attack detection and security assessment
- Implementation guidelines and best practices documentation
- Technical training materials and educational modules
- Peer-reviewed publications advancing the field of secure robotics
- Validated experimental datasets for research community

6. References

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization

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